Classwork 05//Tokenization and Embedding

Student Name: ______________________ **Points:** 30

A Number: ______________________ **Due Date:** Feb 24, 2026

1. In natural language processing, there are different methods of tokenization to prepare as input for deep-learning based methods. They have become more relevant in the context of transformers and Large Language Models (LLM). Do literature review on three such tokenization methods, not discussed in the class that is applicable to transformer-based deep learning for sequence data (it doesn't have to be text-based only). You need to provide discussion that includes method as well as any mathematical formula as necessary with some examples. References must be provided.

2. Embeddings serve as ways to incorporate not only the positional information but also to reduce the dimensionality of the token vector.

   How do embeddings work in the context of multmodal data input (images, videos, text, point cloud, etc.)? Do literature review on embeddings in the context of multimodal inputs. You need to provide discussion that includes method as well as any mathematical formula as necessary with some examples. References must be provided.